MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) Regular Examination March 2026
Fourth Semester
Discipline Specific Course (DSC) -MG4DSCECO201 - QUANTITATIVE ECONOMICS - II
(2024 ADMISSION ONWARDS)

Duration : 2 hours **Maximum Marks: 70**

Scheme for valuation

Part A

Multiple Choice Questions

Answer all questions.

Each question carries 1 mark.

Give 1 mark each to the correct answers of Qn 1-10.

1. d) Individual information
2. c) Positional average
3. b) Both represent probability distributions
4. c) Retail prices paid by consumers
5. c) Positive
6. a) Two parts
7. b) The independent variable
8. a) In data with outliers
9. b) Proportional
10. c) There is a strong positive correlation, but it does not imply causation

Part B

Short Answer

Answer any 10 questions.

Each question carries 3 marks.

11. Simple Bar Diagram shows only one variable using single bars for each category. – 1.5 mark

Multiple Bar Diagram shows two or more related variables using grouped bars for each category. 1.5 mark

12. The **Z-score** tells how many standard deviations a value is above or below the mean.

Uses: To standardize data for comparison, To identify outliers **,** To find probabilities in a normal distribution

Formula

$Z = \frac{X-\mu}{\sigma}$

X = individual value

μ (mu) = mean of the dataset

σ (sigma) = standard deviation

13. Coefficient of Range $= \frac{L-S}{L+S}$

Where:

L = Largest value = 80

S = Smallest value = 20

Calculation:

$$\frac{80-20}{80+20} = \frac{60}{100} = 0.6$$

14.Spearman's Rank Correlation Coefficient (ρ).

Formula:

$$\rho = 1 - \frac{6\sum d^2}{n(n2-1)}$$

Student	**X**	**Y**	**d = X−Y**	**d^2**
1	1	1	0	0
2	2	3	-1	1
3	3	2	1	1
4	4	5	-1	1
5	5	4	1	1

Student	X	Y	d = X−Y	d^2
6	6	6	0	0

$\sum d^2=4$

$$\rho = 1 - \frac{6 \times 4}{6(36-1)} = 1 - 21024 = 1 - 0.114 = 0.886$$

$\rho \approx 0.89 \rightarrow$ Strong positive correlation

Interpretation: The teachers mostly agree in their rankings.

15. Multistage sampling is a method of sampling in which the selection of samples is carried out in two or more stages, using smaller and smaller units at each stage.

Example: Country → State → District → Village → Household

16. Binomial Distribution Formula

$$P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}$$

Where

n = number of trials

x = number of successes

p = probability of success

(1 − p) = probability of failure

17. Mean $= \frac{7+9+11+13}{4} = \frac{40}{4} = 10$

Value (X)	|X − Mean|
7	3
9	1
11	1
13	3

$$\text{Mean Deviation} = \frac{3+1+1+3}{4} = \frac{8}{4} = 2$$

18. r = -0.85 indicates a strong negative correlation between price and demand.

- As price increases**,** demand decreases significantly
- As price decreases**,** demand increases

There is a strong inverse relationship between price and demand, which is consistent with the law of demand in economics.

19. Symmetrical Distribution : Data is evenly distributed on both sides of the center

- Left side = Right side
- Mean = Median = Mode
- Example: Normal distribution

Skewed Distribution :Data is not evenly distributed

- One tail is longer than the other
- Mean, Median, Mode are not equal
- Types:
  - Positive skew → tail on the right
  - Negative skew → tail on the left

20. GDP (Gross Domestic Product) : GDP is recorded yearly.

A moving average helps smooth short-term fluctuations and shows the long-term economic growth trend.

Stock Prices : Stock prices change daily.

A moving average (e.g., 7-day or 30-day) helps identify the trend by reducing daily volatility

21. A finite population is a population that contains a fixed and countable number of elements. The total number of units is limited and known
Example: Number of students in a class or employees in a company

22. Value Liberally
Mean = 15.09
Variance = 64.05
Standard Deviation = 8.00
Coefficient of Variation = 53%

23. Gross Domestic Product (GDP), Inflation Rate (CPI/WPI), Unemployment Rate, Stock Prices, Interest Rates , Exchange Rates etc.. (Any two)

Part C

Short Essay

Answer any 5 questions.

Each question carries 6 marks.

24. Secondary data refers to data that has already been collected, processed, and published by someone else for a purpose different from the current research. Instead of gathering fresh (primary) data, researchers use existing information available in reports, publications, or databases. Secondary data can be broadly classified into published and unpublished sources:

Published Sources :

- Government Publications: Census reports, Economic Surveys, Statistical Abstracts, reports from ministries and departments.
- International Organizations: Reports from organizations like World Bank, International Monetary Fund, United Nations.
- Research Institutions and Universities: Academic journals, working papers, theses, and dissertations.
- Books, Journals, Newspapers, and Magazines: Provide qualitative and quantitative information.
- Websites and Online Databases: Government portals, company websites, and digital archives.

2. Unpublished Sources

- Records maintained by private organizations, firms, and NGOs.
- Personal documents such as diaries, letters, and manuscripts.
- Internal reports, business records, and confidential files.

25. Given Data (arranged in ascending order):

20, 25, 30, 35, 40, 45, 50, 100

n=8

(a) Range

Range=Maximum value −Minimum value

=100−20=80

Range = 80

(b) Quartile Deviation (Q.D.)

Quartile Deviation is:

$$Q_1 = \text{value of } \frac{(n+1)}{4}\text{th item}, \quad Q_3 = \text{value of } \frac{3(n+1)}{4}\text{th item}$$

$$Q_1 = \frac{9}{4} = 2.25\text{th item}$$

$$Q_3 = \frac{27}{4} = 6.75\text{th item}$$

- 2nd item = 25, 3rd item = 30

$$Q_1 = 25 + 0.25(30 - 25) = 25 + 1.25 = 26.25$$

- 6th item = 45, 7th item = 50

$$Q_3 = 45 + 0.75(50 - 45) = 45 + 3.75 = 48.75$$

$$QD = \frac{48.75 - 26.25}{2} = \frac{22.5}{2} = 11.25$$

26. Probability sampling is a sampling technique in which every unit of the population has a **known and non-zero chance** of being selected. Selection is done using random methods, ensuring that the sample is representative and free from personal bias. It is widely used in scientific

and statistical research because it allows valid generalizations about the population.

Major Probability Sampling Methods

1. Simple Random Sampling : Every unit has an equal chance of selection. Selection is done using random numbers, lottery method, or computer-generated randomization.

2. Systematic Sampling : Selection of every kth unit from a list after a random start. k=N/n, where N = population size and n = sample size.

3. Stratified Sampling : Population is divided into homogeneous groups (strata) based on characteristics (e.g., age, income).
4. Cluster Sampling: Population is divided into clusters (usually geographically).

5. Multistage Sampling : Sampling is done in stages using a combination of methods. Example: Selecting districts → villages → households.

27. Less-than cumulative frequency distribution**, so** first convert it into a class frequency table**.**

Class Interval	Cumulative Frequency	Frequency
0–5	3	3
5–10	11	11 − 3 = 8
10–15	16	16 − 11 = 5
15–20	32	32 − 16 = 16
20–25	58	58 − 32 = 26
25–30	70	70 − 58 = 12
30–35	76	76 − 70 = 6
35–40	79	79 − 76 = 3

$$N = 79$$

$$\frac{N}{2} = \frac{79}{2} = 39.5$$

The cumulative frequency just greater than **39.5** is **58**, corresponding to class **20–25**.

Median class = **20–25**

$$\text{Median} = L + \left(\frac{\frac{N}{2} - C}{f} \right) \times h$$

Where:

- $L = 20$ (lower limit of median class)
- $C = 32$ (cumulative frequency before median class)
- $f = 26$ (frequency of median class)
- $h = 5$ (class width)

$$\text{Median} = 20 + \left(\frac{39.5 - 32}{26} \right) \times 5$$

$$= 20 + \left(\frac{7.5}{26} \right) \times 5$$

$$= 20 + 1.44 \approx 21.44$$

Median = 21.44.

28. The normal distribution (also called the Gaussian distribution) is one of the most important continuous probability distributions in statistics. It is widely used in natural and social sciences due to its mathematical properties and its ability to approximate many real-life phenomena.

1. Bell-Shaped and Symmetrical
2. Mean = Median = Mode
3. Total Area Equals One
4. Asymptotic Nature
5. Defined by Two Parameters: The shape of the normal distribution is completely determined by: Mean (μ) – determines the location (center) Standard Deviation (σ) – determines the spread (width)
6. Points of Inflection: The curve changes its curvature at two points called points of inflection: Located at μ ± σ These points mark where the curve shifts from concave to convex.

7. Empirical Rule (68–95–99.7 Rule): A key property of the normal distribution. About 68% of values lie within ±1σ . About 95% lie within ±2σ . About 99.7% lie within ±3σ
8. The normal distribution is a continuous probability distribution:
9. Skewness and Kurtosis
10. Standard Normal Distribution

29.

Commodity	(P0)	(Q0)	(P1)	(Q1)
A	3	10	5	12
B	6	5	7	4
C	8	4	9	6

Fisher's Index:

$$F = \sqrt{L \times P}$$

Laspeyres Price Index (L)

$$L = \frac{\sum P_1 Q_0}{\sum P_0 Q_0} \times 100$$

$$\sum P_1 Q_0 = (5 \times 10) + (7 \times 5) + (9 \times 4) = 50 + 35 + 36 = 121$$

$$\sum P_0 Q_0 = (3 \times 10) + (6 \times 5) + (8 \times 4) = 30 + 30 + 32 = 92$$

From above:

$$L = \frac{121}{92} \times 100 = 131.52$$

Paasche Price Index (P)

$$P = \frac{\sum P_1 Q_1}{\sum P_0 Q_1} \times 100$$

$$\sum P_1 Q_1 = (5 \times 12) + (7 \times 4) + (9 \times 6) = 60 + 28 + 54 = 142$$

$$\sum P_0 Q_1 = (3 \times 12) + (6 \times 4) + (8 \times 6) = 36 + 24 + 48 = 108$$

$$P = \frac{142}{108} \times 100 = 131.48$$

Fisher's Index

$$F = \sqrt{131.52 \downarrow 31.48} \approx 131.5$$

(ii) Marshall–Edgeworth Index

$$ME = \frac{\sum P_1 (Q_0 + Q_1)}{\sum P_0 (Q_0 + Q_1)} \times 100$$

Calculate:

$$Q_0 + Q_1 :$$

Commodity	(Qo + Q1)
A	22
B	9
C	10

$$\sum P_1 (Q_0 + Q_1) = (5 \times 22) + (7 \times 9) + (9 \times 10) = 110 + 63 + 90 = 263$$

$$\sum P_0 (Q_0 + Q_1) = (3 \times 22) + (6 \times 9) + (8 \times 10) = 66 + 54 + 80 = 200$$

$$ME = \frac{263}{200} \times 100 = 131.5$$

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30. Given

$$\bar{x} = 65, \quad \bar{y} = 67, \quad \sigma_x = 2.5, \quad \sigma_y = 3.5, \quad r = 0.8$$

a. **Regression equation of Y on X**

$Y = a+bx$

$b = r(\sigma x/\sigma y)$

$= 0.8 * 3.5 / 2.5 = 1.12$

$a = \bar{y} - b\bar{x}$

$= 67 - (1.12*65) = -5.8$

$Y = -5.8 + 1.12X$

b. **Regression equation of X on Y**

$X = a' + b'y$

$b' = r(\sigma y/\sigma x)$

$= 0.8*2.5/3.5 = 0.57$

$a' = \bar{x} - b\bar{y}$

$= 65-(0.57*67) = 26.81$

$X = 26.81+0.57y$

Estimate Y when X= 60

$Y = -5.8 + 1.12X$

$Y = -5.8+(1.12*60) = 61.54$

Estimate X when Y= 70

$X = 26.81+0.57y$

$X = 26.81+(0.57*70)=66.71$